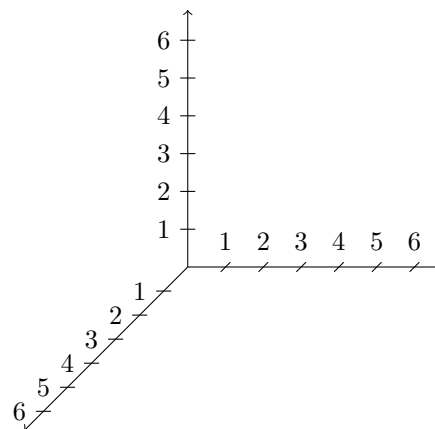


1 Three Dimensional Coordinate Systems and Vectors

1.1 Three Dimensional Coordinate Systems



We describe 3-dimensional space by picking an origin, and three perpendicular axis, namely x , y , and z -axis. these must follow the **right hand rule**.

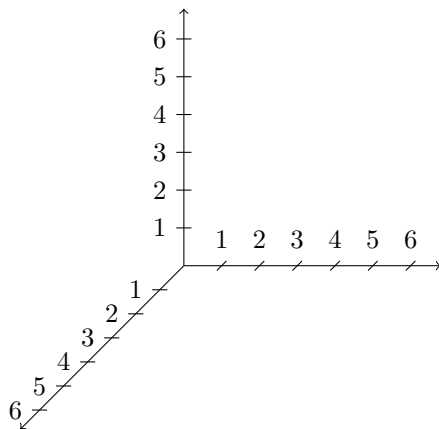
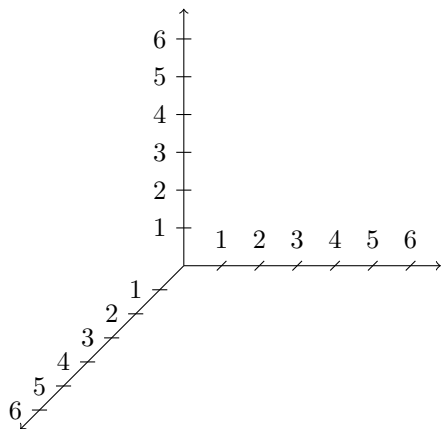
For any point in 3-dimentional space, these three axis give us a way to describe where that point is, namely a magnitude in each of the three directions of the axes. We call this the coordinate of the point.

Give an example of plotting a point in 3-space.

1.1.1 Surfaces and Solids

Example 1.1.1 Describe the surfaces defined by the following equations.

- $z = 3$
- $y = 5$

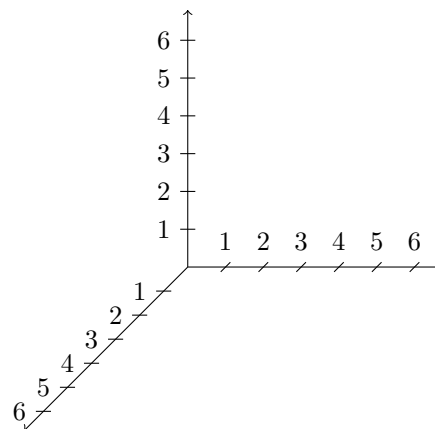


Mention that in 2-dimensional geometry, graphs of equations involving x and y are curves in the plane, while here, graphs of equations involving x , y , and z are surfaces.

Example 1.1.2 Describe the set of points which satisfy the two equations

$$x^2 + y^2 = 1 \quad \text{and} \quad z = 3.$$

If the restriction that $z = 3$ had not been given, we would have a cylinder. Draw out what this would look like with no restriction, and with the restriction $1 \leq z \leq 4$.



Definition 1.1.3 (Distance Formula) Given two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$, the distance between them is

$$d(P_1, P_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}.$$

Example 1.1.4 Find the distance between the two points $P_1(2, -1, 7)$ and $P_2(1, -3, 5)$.

Definition 1.1.5 (Equation of a Sphere) An equation of the sphere with center $P(a, b, c)$ and radius $r \geq 0$ is given by

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2.$$

Example 1.1.6 Find the equation of the sphere with center $(1, 1, 1)$ which passes through $(2, -1, 2)$.

Example 1.1.7 Show that the equation

$$x^2 + y^2 + z^2 + 4x - 6y = 0$$

is the equation of a sphere, and find its center and radius.